EXP.NO: 6	IPC using Shared Memory
DATE: 20/02/2026	

AIM:

To implement a C program that facilitates communication between two independent processes (a Sender and a Receiver) using System V Shared Memory. The objective is to allow the Sender to write a string into a shared memory segment and the Receiver to read and display that same string.

ALGORITHM:

1. Identify the Segment

Both processes define a unique **Key** (e.g., `1234`). This acts as a common ID so they can find the same memory location.

2. Create / Access (`shmget`)

The **Sender** creates a 1024-byte memory segment using `shmget` with the `IPC_CREAT` flag.

The **Receiver** uses the same key to locate the existing segment.

3. Attach to Address Space (`shmat`)

Both processes "map" the shared memory into their own memory space. This returns a pointer (`str`), allowing them to interact with the shared memory as if it were a local variable.

4. Data Exchange

Sender: Takes user input via `scanf` and writes it to the memory pointed to by `str`.

Receiver: Reads the string directly from the same pointer `str` and prints it.

5. Detach (`shmdt`)

The **Receiver** detaches from the memory segment once the reading is complete to ensure clean termination.

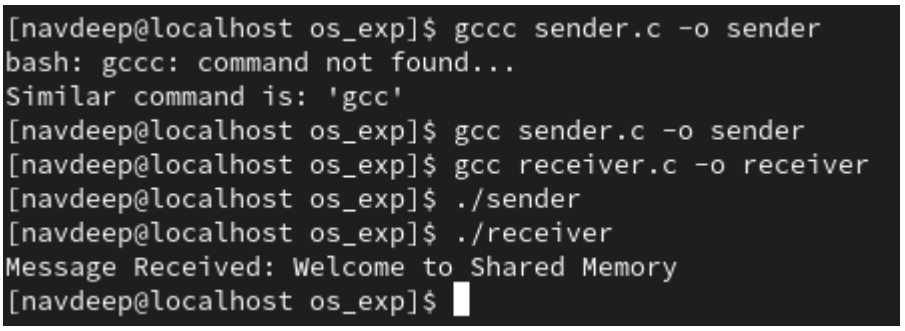
CODE:

```
// sender

#include <stdio.h>      // for printf, scanf
#include <sys/ipc.h>    // for key_t
```

```
#include <sys/shm.h>    // for shared memory functions
#include <string.h>     // for string operations

int main() {
   key_t key = 1234;   // Unique key to identify shared memory
   int shmid;          // Variable to store shared memory ID

   // Create shared memory segment of size 1024 bytes
   // 0666 = read/write permission for all users
   // IPC_CREAT = create memory if it doesn't exist
   shmid = shmget(key, 1024, 0666 | IPC_CREAT); // Create

   // Attach shared memory to this process
   // Returns pointer to the shared memory
   char *str = (char*) shmat(shmid, (void*)0, 0); // Attach

   printf("Enter message: ");

   // Read a full line (including spaces) and store directly in shared memory
   scanf(" %[^\n]", str);

   // Print the data written into shared memory
   printf("Data written to shared memory: %s\n", str);

   return 0;   // End program
}

//receiver

#include <stdio.h>      // for printf
#include <sys/ipc.h>    // for key_t
#include <sys/shm.h>    // for shared memory functions

int main() {
   key_t key = 1234;   // Same key used by sender
   int shmid;          // Variable to store shared memory ID

   // Access existing shared memory
   // No IPC_CREAT → it must already exist
   shmid = shmget(key, 1024, 0666); // Get

   // Attach shared memory to this process
   char *str = (char*) shmat(shmid, (void*)0, 0); // Attach
```

```
    // Read and print data from shared memory
    printf("Data read from shared memory: %s\n", str);

    // Detach shared memory after use
    shmdt(str);

    return 0;   // End program
}
```

OUTPUT:



RESULT:

The program successfully simulates process synchronization. The producer is restricted from adding items when the buffer is full (e=0), and the consumer is restricted from removing items when the buffer is empty (f=0), maintaining data integrity within the shared resource.